

EVALUATING AND MITIGATING BIAS IN SPOTIFY'S RECOMMENDATION SYSTEM: AN AI-DRIVEN APPROACH USING RANDOM FOREST AND CLUSTER ANALYSIS



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Abstract:

Music recommendation systems have a key role in shaping how users discover songs and artists gain visibility on streaming platforms. Current recommendation algorithms prioritize historical popularity and engagement metrics, which can reinforce popularity bias and limit the discovery of smaller artists, preventing them from reaching new listeners and achieving success in the industry. This study investigates whether a fairness-constrained AI framework can improve unknown artists' visibility and reduce algorithmic bias in music recommendations. Using RStudio and a publicly available Spotify music listening dataset, we apply AI-based methods, including principal component analysis (PCA), cluster analysis, and random forest modeling, to simulate recommendation strategies. A random forest model predicting popularity from audio features alone showed low explanatory power ($R^2 \approx 0.17$), indicating that musical characteristics explain only a small portion of popularity. This suggests that external platform-driven factors, such as historical engagement and exposure, play a dominant role in shaping song popularity. Our approach reframes popularity as an ethical design choice rather than a calculated metric. Original rankings included 0% small artists in the top 1,000 recommendations, while the AI-based model increased representation to approximately 9.8%, with further gains to 13.1% under fairness adjustments and only a slight decrease in average popularity. This model can be applied to individual listeners to recommend unknown songs or artists based on their prior listening history, without the bias of the plays/skip popularity score. Previously "unpopular" songs now gain visibility under this framework, increasing the representation of smaller artists. This recommendation algorithm could be used in the future on other platforms such as YouTube, Netflix, or TikTok, allowing emerging creators to gain greater visibility and audience reach. Furthermore, these findings call into question whether prioritizing popular artists is an intentional choice or simply a product of user listening history, complicating the ethical justification of such practices.

Background Information:

- Music recommendation systems shape how users discover songs and how artists gain visibility on platforms like Spotify.
- Current systems rely on engagement metrics (e.g., streams, playlists), reinforcing popularity bias.
- This creates limited exposure for smaller or emerging artists.
- Prior research shows recommendation systems can embed bias through feedback loops.
- Feature-based approaches (using audio characteristics like danceability, energy, etc.) offer a more balanced alternative.
- This study evaluates whether clustering + random forest models can reduce popularity bias by generating recommendations based on audio features rather than engagement.

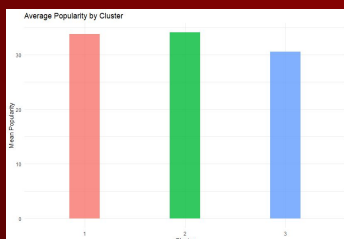


Figure 1: Average popularity by cluster with confidence intervals. The results show that Cluster 2 has the highest mean popularity, followed by Cluster 1 and Cluster 3.

Method/Materials:

- **Spotify Tracks Dataset**
 - Retrieved from Kaggle.com.
 - Includes a numerical popularity metric.
 - Numerical audio features include danceability, energy, valence, acousticness, instrumentalness, liveness, speechiness, and popularity.
- **RStudio (v.4.2.1)**
 - Data cleaning was performed to remove duplicates based on their unique track_id.
 - Audio features standardized using z-score normalization.
 - K-means clustering: to group similar songs.
 - ANOVA: to determine whether average song popularity differs significantly across clusters.
 - Random forest: to predict popularity based only on audio features to create the synthetic popularity.
 - PCA: to reduce data complexity and visualize how songs cluster.
 - Packages used: readr, ggplot2, randomForest, dplyr.

Results:

- 89,741 unique tracks remain after cleaning (21% of tracks were removed).
- Sample of 15,000 songs was used.
- Three distinct song clusters identified based on audio features:
 - Cluster 1: High Energy & Tempo
 - Cluster 2: High Danceability & Valence (Most Popular)
 - Cluster 3: Acoustic & Instrumental (Least Popular)
- Differences in popularity were statistically significant.
- Random forest model showed low predictive power:
 - $R^2 \approx 0.168 \rightarrow$ audio features explain only a small portion of popularity.
 - RMSE = 18.85
- Original ranking included 0% small artists in top 1,000 songs.
- Synthetic (audio-based) ranking increased small artist representation to 7%-10% in top 1,000 songs.
- Hard fairness constraint achieved 50% representation of small artists in top 1,000, but significantly reduced popularity.
- Fairness-adjusted model increased representation of small artists in top 1,000 to 13%, while only slightly decreasing average popularity.

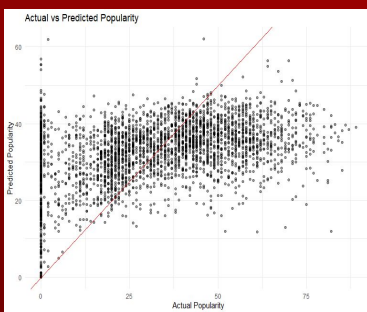


Figure 6: Actual versus predicted popularity. The wide dispersion of points around the diagonal line indicates weak predictive accuracy, reinforcing that audio features alone do not strongly determine popularity.

PCA Visualization of 3 Clusters

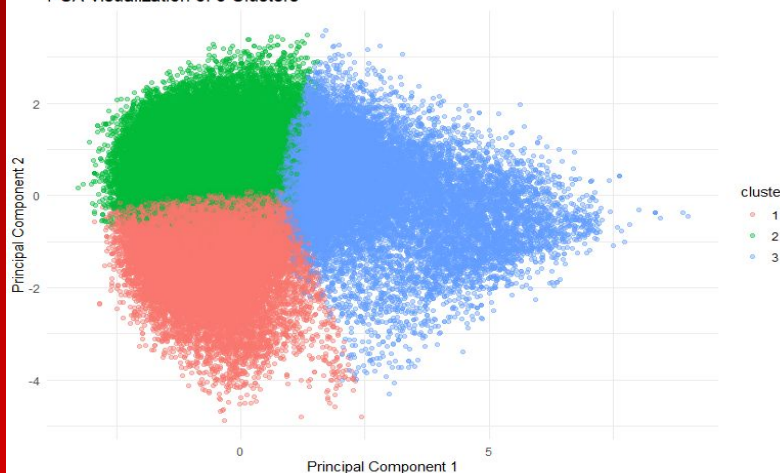


Figure 4: PCA visualization of clustered songs. The first principal component (PC1) captures a spectrum from acoustic to energetic tracks, while the second component (PC2) reflects a transition from darker, less danceable songs to more positive and danceable compositions. The clusters remain well-separated, confirming the robustness of the segmentation.

Random Forest Variable Importance

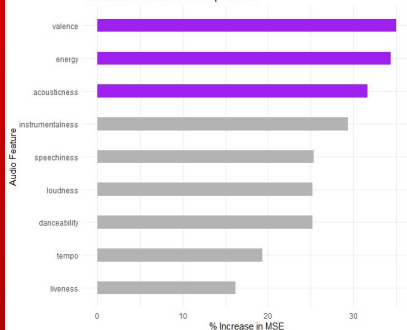


Figure 7: Random Forest variable importance. Valence, energy, and acousticness contribute most to predictive performance, while features such as tempo and liveness have lower importance.

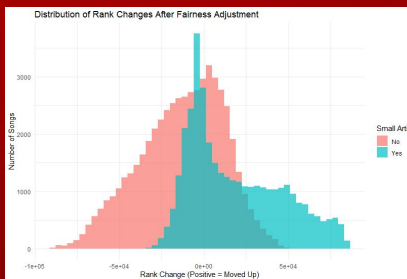


Figure 8: Distribution of rank changes after fairness adjustment. Most songs experience minimal shifts in ranking position, as indicated by the concentration of values near zero.

Discussion/Conclusion:

- Audio features explain only 17% of the variation in popularity.
 - Suggests that artist reputation, marketing exposure, playlist placement, and platform algorithms significantly influence popularity scores.
- Original Spotify recommendation system showed strong popularity bias, as no small artists appeared in top 1,000 songs.
- Fairness improvements came with minimal loss in overall recommendation quality.
- Results highlight a tradeoff between engagement and ethical responsibility.
- Creates an opportunity for streaming platforms to incorporate fairness-aware mechanisms into recommendation systems.
- Can be applied to other platforms that provide recommendations: Netflix, Youtube, TikTok, etc.

Credit Authorship Contribution Statement:

Blankenbiller, M.: Editing, Rstudio coding, Figure 4, Figure 8, writing- Abstract, Results, Discussion, Conclusion;
Cascone, R.: Editing, Rstudio coding, Figure 1, Figure 6, writing- Abstract, Methods, Results, References; **Dwyer, K.:** editing, Rstudio coding, Figure 7, writing- Abstract, Methods, Results; **Marsellos, A.E.:** supervision, guidance, editing; **Yetgin, E.:** supervision, guidance, editing

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USING ARTIFICIAL INTELLIGENCE TO IDENTIFY KEY PERFORMANCE DRIVERS IN MAJOR LEAGUE BASEBALL



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Abstract:

Identifying the metrics that drive success in Major League Baseball remains a key challenge in sports analytics. This study applies artificial intelligence (AI) to uncover key performance indicators and to support data-driven decision-making using MLB Statcast data in professional baseball. Using five years of Statcast data, we applied AI-driven methods, including K-means clustering, multiple regression, and Random Forest models, to identify performance patterns and predict player success in a professional sports context. K-means clustering identified distinct groups of hitters, where high-value players combined higher on-base performance with more selective swing behavior, while lower-value players showed weaker discipline and production.

Regression results explained 55.5% of the variance in players' Wins Above Replacement (WAR) ($R^2 = 0.555$, $p < .001$), indicating predictive power, though a significant portion of performance remained unexplained. Slugging percentage emerged as the strongest predictor, followed by contact on pitches outside the strike zone, while poor swing decisions negatively impacted performance. However, because a significant portion of performance remains unexplained, overreliance on AI may overlook contextual factors such as player health and situational performance, emphasizing the need to use these models as decision-support tools rather than replacements for human judgment. These findings demonstrate how AI can help MLB teams identify key performance drivers and improve strategic, data-driven decision-making, yet human oversight is still needed to account for unencoded contextual and situational factors not captured in the model.

Background Information:

Statcast provides detailed, high-frequency data on player behavior, including swing decisions, contact rates, and power metrics. These variables allow for deeper insights into key aspects of hitting performance, such as plate discipline and power production.

Artificial intelligence (AI) methods, such as clustering and machine learning, have emerged as powerful tools for identifying patterns in large datasets (Kuhn, 2008; Liaw & Wiener, 2002). These approaches can uncover complex relationships between performance variables that are not easily detected using traditional statistical methods (Kuhn, 2008).

This study applies AI-driven techniques to determine which offensive metrics most strongly influence player value, measured by Wins Above Replacement (WAR), and evaluates how these tools can support data-driven decision-making in professional baseball.

Methods/Data:

- **Data Source:** MLB Statcast data (2021–2023)
 - 122 players, 176 performance variables
 - Focus on plate discipline, contact ability, and power metrics
- **Analytical Tools:** RStudio (data cleaning, modeling, visualization)
- **K-Means Clustering**
 - Variables: OBP, WAR, Z-Swing%
 - Purpose: Identify distinct hitter profiles
- **Random Forest Model**
 - Used to determine variable importance
 - Dataset split into training/testing sets
 - Identified most influential predictors of offensive success
- **Multiple Linear Regression:** Evaluated using R^2 and significance testing
 - **Dependent variable:** WAR
 - **Independent variables:** SLG (Slugging %), O-Swing%, O-Contact%, Z-Swing%, Z-Contact%

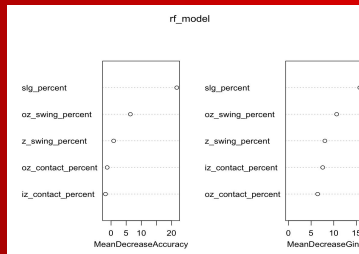


Figure 1: 2021 Random Forest model

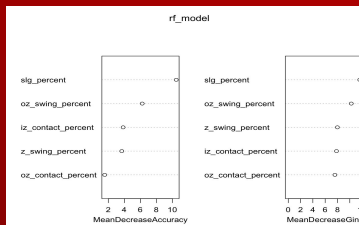


Figure 2: 2022 Random Forest model

Credit Authorship Contribution Statement:

Buchanan, K.: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data Curation, Writing, RStudio coding; **Kirkpatrick, S.:** Methodology, Formal analysis, Investigation, Review & Editing; **Stoeckel, F.:** Conceptualization, Methodology, Software, Visualization, Graphics, Review & Editing; **Marsellos, A.E.:** Supervision, Project administration, Guidance, Editing; **Yetgin, E.:** Supervision, Project administration, Guidance, Editing.

Results:

Clustering Findings

- Four distinct hitter profiles identified
- High-value players: Higher OBP, better plate discipline (selective swinging)
- Lower-value players: Poor discipline, lower offensive production

Regression Results

- Model significance: $F(5,455) = 113.68$, $p < .001$
- Explained variance: $R^2 = 0.555$

Key Predictors

- Slugging % (SLG) → strongest positive predictor
- O-Contact% → positive effect
- O-Swing% → negative effect

Simplified Model

- Similar performance ($R^2 = 0.553$)
- Confirms: Power hitting and plate discipline are primary drivers of success

Overall Insight

Players who hit for power, avoid chasing bad pitches, still make contact when they do
→ produce the highest value (WAR)

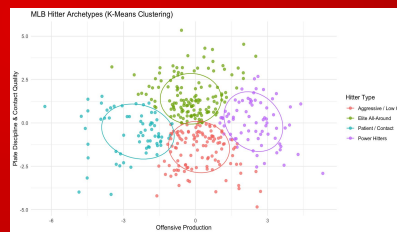


Figure 3: 2021-2023 k-means clustering plate discipline & contact quality by offensive production clustered by hitter type

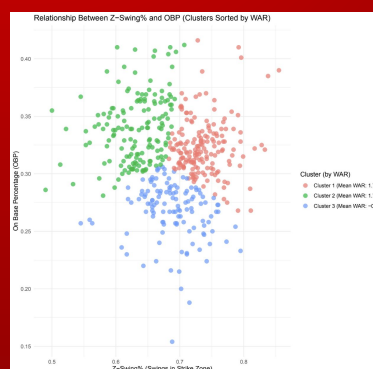


Figure 4: 2021-2023 k-means clustering relationship between Z-Swing% and OBP clustered by WAR

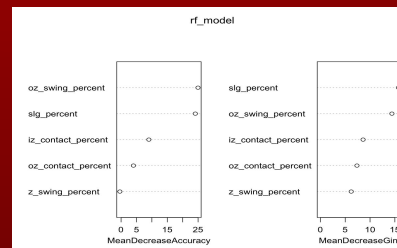


Figure 5: 2023 Random forest model

Discussion:

Artificial intelligence is playing a growing role in identifying key MLB performance drivers. Across models, **slugging percentage (SLG)** consistently emerged as the strongest predictor of offensive success, reinforcing the value of power hitting.

Plate discipline was also critical: lower **O-Swing%** and higher **O-Contact%** significantly improved performance. Successful hitters combine power with selectivity and adaptability at the plate.

K-means clustering revealed that high-value players blend strong on-base skills with disciplined swing behavior. These findings support modern analytics focused on efficiency and smart decision-making.

The models explained only **55% of WAR variance**, leaving nearly half of player performance unexplained. Factors like health, defense, and situational play likely account for the rest. AI excels at pattern detection but cannot replace human judgment.

Key Takeaway: AI should support, not replace, scouting and coaching expertise.

Conclusion:

This study demonstrates that AI effectively uncovers key performance drivers in MLB using Statcast data. **Slugging percentage**, plate discipline, and outside-zone contact ability were the strongest predictors of offensive success.

However, with **45% of performance variability unexplained**, statistical models have clear limits. AI analytics should complement traditional evaluation methods.

Future Directions:

- Include defensive metrics, injury history, and situational data
 - Explore advanced modeling techniques
 - Integrate AI with human scouting and coaching
- Combining data-driven insights with human expertise will help teams gain a competitive edge.

Acknowledgements:

We would like to thank Dr. Antonios E. Marsellos & Dr. Emre Yetgin and the Norm Brodsky College of Business at Rider University for providing us with the necessary tools to conduct this research.

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AI-DRIVEN ANALYSIS OF GLOBAL EMISSIONS: INSIGHTS AND ETHICAL LIMITS FOR SUSTAINABLE DECISION-MAKING



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Abstract:

Artificial Intelligence (AI) is increasingly integrated into business decision-making, particularly in environmental analysis. While AI systems can model complex relationships in environmental data, they lack the capacity to independently interpret ethical concepts such as what constitutes "sustainable" outcomes. This study investigates how AI model outputs are influenced by human-defined ethical frameworks used to classify and evaluate sustainability.

Using global datasets on greenhouse gas emissions, trade activity, and transportation, we applied K-means clustering and random forest models in RStudio to identify patterns and predict sustainability-related outcomes. K-means clustering ($k = 3$) grouped 148 countries into low ($n = 61$), intermediate ($n = 40$), and high ($n = 47$) profiles based on GDP, emissions, and mobility. A random forest regression model demonstrated moderate predictive power ($R^2 = 0.305$), identifying GDP per capita as the strongest predictor of emissions. A classification model achieved 91.67% accuracy ($k = 0.87$) in recovering clusters. An animated choropleth map indicated increasing per-capita emissions over time with shifting geographic patterns.

The findings indicate that although AI methods effectively identified patterns and improved the accessibility of environmental data, model outputs depended heavily on how sustainability was defined and labeled. This demonstrates that AI systems were not neutral tools but were shaped by the ethical assumptions embedded in their design. While AI enhances analytical capabilities, it could not determine what outcomes were ethically acceptable without human input, reinforcing the need for human oversight in sustainability decision-making.

Background

Information:

- Artificial Intelligence (AI) is increasingly integrated into business decision making, particularly in environmental analysis
- Corporations need to consider different coinciding factors when evaluating global sustainability and how it affects their operations
- With AI rapidly being used in business operations, the balance between AI-decision making and human-based decision making for business operations is an evolving point of discussion.

- This study uses global datasets on greenhouse gas emissions, trade activity measured through GDP, and transportation measured by air passenger mobility. These are used to identify patterns and predict sustainability-related outcomes

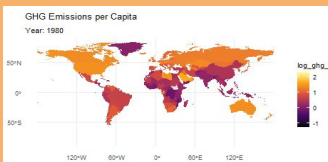


Figure 1: Animated choropleth plot visualizing global emission. Higher emissions are concentrated in more developed or industrialized countries

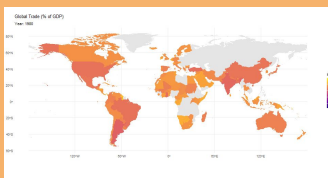


Figure 2: Animated choropleth plot visualizing global trade.

Method/Materials:

- World Development Indicators (WDI) Databank**
 - GHG, Air Passenger, and GDP data from 148 countries
 - Data from 2022 across all datasets for each country
- Natural Earth Global Country Shapefile**
 - Provided geographic boundary data and coordinate reference system
 - Spatial join key to link country data into mappable polygons
- RStudio (v.4.2.1)**
 - Extrapolated data from WDI
 - Merged datasets (GDP, GHG, and Air Passengers)
 - Packages used: ggplot2, maturalearth, viridis, cluster, caret, randomForest, etc.
 - K-Means Clustering (elbow method, silhouette analysis)
 - Random Forest Analysis (regression and classification)

Results:

Choropleth & Animated Map Findings

- Developed regions consistently showed higher emissions and mobility.
- Developing regions showed lower values across all indicators.
- Animated map showed emissions changing over time alongside economic growth and industrialization.

K-Means Clustering Results

- Elbow and silhouette methods supported $k = 3$ as the optimal number of clusters.
- Clusters reflected low, moderate, and high development/emissions profiles.

Random Forest Regression Results

- GDP per capita was the strongest predictor of GHG emissions.
- Model explained a modest amount of variation:
 - $R^2 = 0.305$
 - $MSE = 0.144$

Random Forest Classification Results

- Model accurately predicted cluster membership:
 - Test Accuracy = 91.67%
 - Kappa = 0.8708
- GDP per capita had the highest variable importance.

Overall Insight

Higher economic development and mobility were generally associated with higher per-capita emissions.

Global Development, Mobility, and Emissions (2022)

Per Capita Indicators (Log Scale)

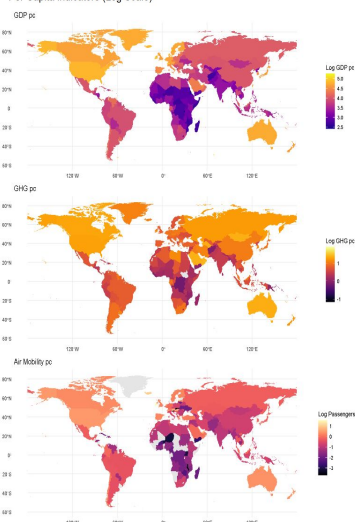


Figure 3: Initial choropleth plot visualizing global distribution of GDP, emissions, and mobility

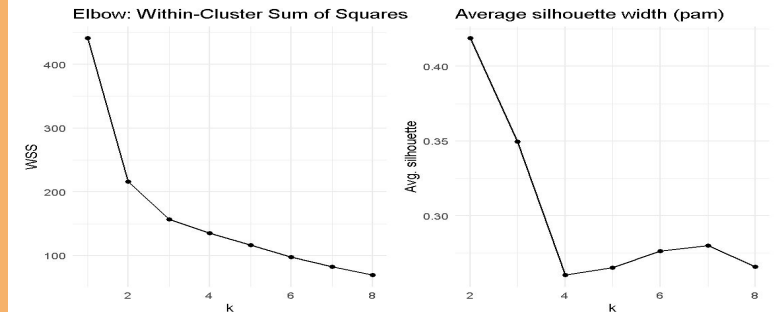


Figure 4: Elbow graph Used to determine the optimal number of clusters (k)

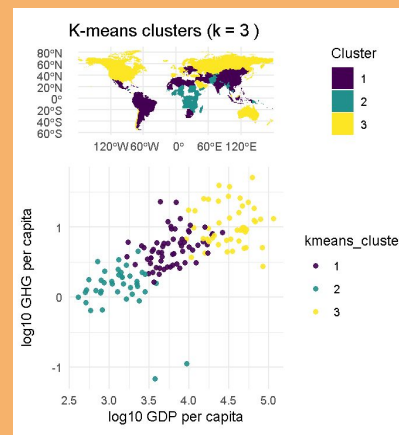


Figure 5: K-means clustering graphs visualizing the countries grouped into 3 clusters. Cluster 2 has the lowest levels across the 3 predictors, followed by Cluster 1 then Cluster 3.

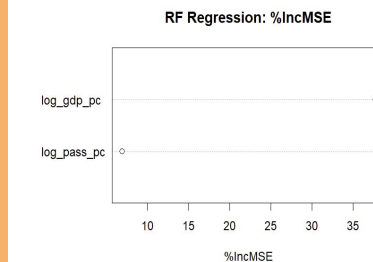


Figure 6: Random Forest Regression graph using %IncMSE to determine which variables hold the most importance in making accurate predictions.

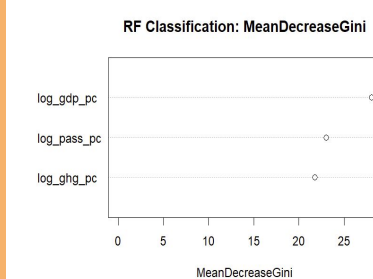


Figure 7: Random Forest Classification graph using MeanDecreaseGini. The model predicts which cluster each country belongs to using GDP, GHG, and passenger mobility as inputs

Discussion:

- AI effectively identified large-scale sustainability patterns across countries.
 - Less effective at explaining why environmental patterns exist.
- Economic development was strongly associated with higher emissions, but the relationship was not universal.
- Policy, energy systems, and efficiency likely influence environmental outcomes beyond GDP alone.
- Strong classification performance showed AI's ability to organize countries into sustainability profiles.
- Sustainability classifications still depended on human-selected variables and definitions.
- Regression results highlighted a key limitation of AI models:
 - Identifying correlations does not mean identifying causation.
- Regression model explained only 30% of emissions variation, showing limitations of purely data-driven analysis.

Conclusion:

- Human judgment remains necessary when interpreting sustainability trade-offs, ethical concerns, and business implications.
- Many sustainability factors cannot be fully captured through numerical variables alone.
- AI models are most valuable as decision-support tools rather than replacements for human decision-making.
- AI can help businesses visualize, organize, and analyze complex global sustainability data.

Acknowledgements

We would like to thank Dr. Antonio E. Marsellos & Dr. Emre Yetgin and the Norm Brodsky College of Business at Rider University for providing us with the necessary tools to conduct this research.

Credit Authorship Contribution Statement:

Burgos, J.: Editing, Rstudio coding, Figure 6, Figure 7, Abstract, Methods, Results, Discussion, Conclusion; **Desilva, A.:** Editing, Rstudio coding, Figure 4, Figure 5, Abstract, Methods, Results, Discussion; **Wagner, J.:** Editing, Rstudio coding, Figure 1, Figure 2, Figure 3, Abstract, Results, Discussion; **Marsellos, A.E.:** supervision, Rstudio coding, guidance; **Yetgin, E.:** supervision, Rstudio coding, guidance

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FORECASTING FILM INDUSTRY GROWTH: ECONOMIC IMPACT AND AI-DRIVEN DECISION SUPPORT IN NEW JERSEY



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Abstract

This study evaluates the economic impact of New Jersey's expanding film and television industry using forecasting, comparative state analysis, and applied artificial intelligence (AI). The project examines how production growth, supported by New Jersey's 2018 film tax incentive policy and major studio investment, is associated with employment, wages, and industry establishments. We analyze a 20-state comparative quarterly time series dataset covering 2006 through 2024. The dataset combines Bureau of Labor Statistics Quarterly Census of Employment and Wages data for NAICS 5121, the classification for motion picture and video industries, with state GDP, unemployment, inflation, production counts, tax incentives, and film spending data. New Jersey is compared with 10 incentive states and 10 lower-activity non-incentive states. An interview we conducted with an assistant director highlights how New Jersey's tax incentives helped attract major studio investment, including Netflix's planned Fort Monmouth project, and strengthen long-term production infrastructure. Trend analysis shows steady wage growth in New Jersey's industry, with average weekly wages increasing by approximately \$54.64 per year ($R^2 = 0.79$) and total annual wages increasing by approximately \$15.76 million per year ($R^2 = 0.80$). Post-2018 models indicate stronger industry development, including higher weekly wages and a shift from declining establishments before 2018 to increasing establishments afterward. Forecasting projects New Jersey production spending to increase from approximately \$874.7 million in 2025 to \$1.12 billion by 2030, supported by a regression model that explains substantial variation in spending ($R^2 = 0.83$). The study also introduces Location Scout, an AI platform we created to identify high-potential filming and development areas in New Jersey. By translating forecasts into location-based recommendations, Location Scout turns economic modeling into a practical decision-support tool for sustainable industry growth and regional development.



Background Information

New Jersey's 2018 film tax incentive expansion made the state a highly competitive production hub. Beyond direct hiring, this activity generates massive economic spillovers for local vendors, hospitality, and municipal services.

Rapid growth often concentrates exclusively in already-established filming hubs. As a result, newer, officially trained Film Ready Communities need to gain visibility with major producers.

NJ officially trains and certifies towns to be film production ready. These communities are equipped with streamlined permitting, dedicated local liaisons, and the exact infrastructure needed to support major filming logistics.

NJ Film Ready Location Scout acts as a sustainability-focused decision-support tool. Using ethical AI design, it connects production needs directly to community assets, ensuring overlooked towns have a fair chance to attract statewide film spending.

Methods

Data: 20-state film industry dataset, 2006–2024, with 380 state-year records and 7,600 data points

Source: BLS QCEW NAICS 5121 film and television industry data

Measures: Employment, total wages, average weekly wages, establishments

Controls: GDP, unemployment, population, population density

Analysis: Interrupted time-series, state comparison, synthetic control, forecasting

AI Output: NJ Film Ready Location Scout for ethical and sustainable location recommendations

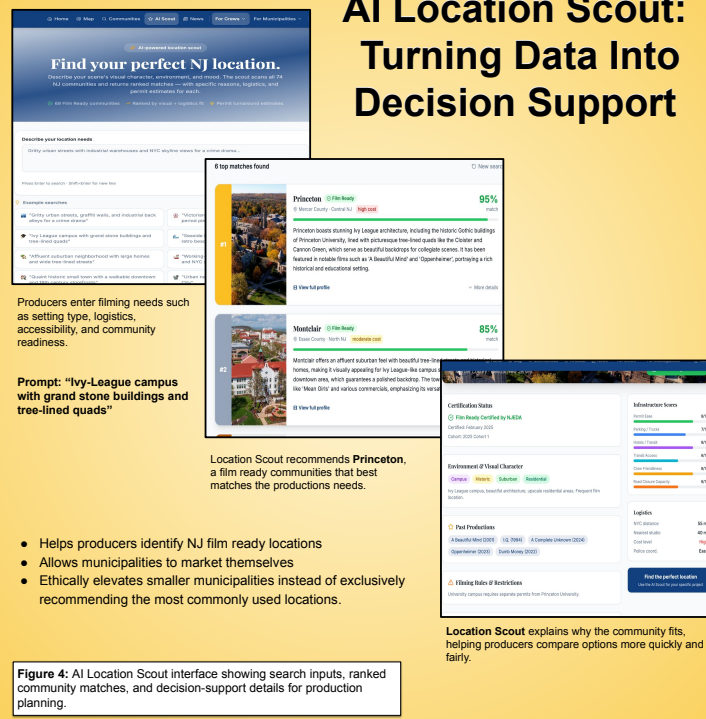
Results

- Average weekly wages increased by \$277 after 2018 and \$349 after 2020 ($R^2 \approx 0.78$).
- Establishments increased by 64 after 2018 and 162 after 2020 ($R^2 \approx 0.92$).
- Total wages increased by approximately \$14.11M after 2018 ($R^2 \approx 0.71$).
- Synthetic control showed NJ's post-2018 weekly wage was about \$301 higher than Synthetic NJ ($RMSPE \approx 65.55$).
- Forecasting projects NJ film spending from \$874.7M in 2025 to \$1.12B by 2030 ($R^2 \approx 0.83$).
- Employment shifted from +34.6 jobs per year before 2018 to -156.6 jobs per year after 2018 ($R^2 \approx 0.23$), showing a more volatile pattern than wages and establishments.

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AI Location Scout: Turning Data Into Decision Support



AI Location Scout Development

- Built as an AI recommendation system that combines a pre-trained LLM with rule-based filters and similarity scoring.
- Google Maps API token used for live geocoding, location searches, and measuring distances to key infrastructure.
- Trained on New Jersey film industry terms, local infrastructure stats, and official Film Ready certification data.
- Uses similarity scoring to Generate a match percentage by comparing a producer's exact needs (like budget, crew size, and setting) directly against a community's actual attributes.
- Programmed to prevent algorithmic bias by ignoring how "popular" a location historically is, ensuring smaller, overlooked towns get a fair chance to be recommended.

Discussion

- The rapid addition of new establishments proves the state's film industry is becoming a permanent ecosystem.
- A First Assistant Director provided real world context for this project, stating the tax incentive is "99% the reason" productions now choose New Jersey (student's interview). He also confirmed that while direct crew jobs fluctuate, local restaurants and hotels see economic spillovers.
- This exact gap motivated the creation of Location Scout. To ensure the projected \$1.12 billion in film spending translates into steady jobs, our ethical AI actively matches productions with smaller, newly certified communities. This democratizes visibility and influences statewide economic development.
- Job creation was temporarily delayed by global shocks like COVID-19 and recent labor strikes, the state's financial data proves the industry remained resilient despite these short-term disruptions.

Conclusion

- Models confirm structural, long-term growth in New Jersey's film wages and establishments, projecting state production spending will reach \$1.12 billion by 2030.
- Applied AI Solution: Location Scout translates these findings into a practical decision-support platform for producers, municipalities, developers, and policymakers.
- Equitable Ecosystem: By prioritizing ethical AI design, the tool prevents algorithmic bias and democratizes visibility for overlooked towns, fostering an accessible and sustainable ecosystem for statewide job creation.

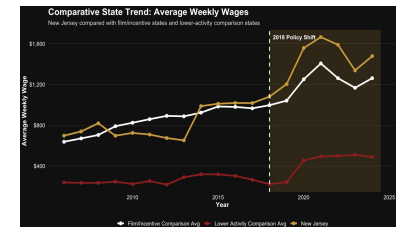


Figure 1: Comparative State Trend: Average Weekly Wages

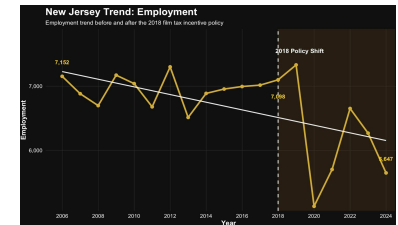


Figure 2: New Jersey Employment Trend

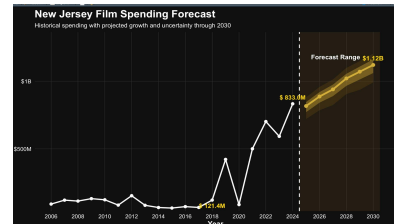


Figure 3: New Jersey Film Spending Forecast

Credit Authorship Contribution Statement

Bacchetta, D.; Lockhart, C.; **McCall Jr, P.;** Implementation, Writing, RStudio coding, Poster design.
Marsellos, A.E.; Supervision, Project administration, Guidance, Editing.
Yetgin, E.; Supervision, Project administration, Guidance, Editing.

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